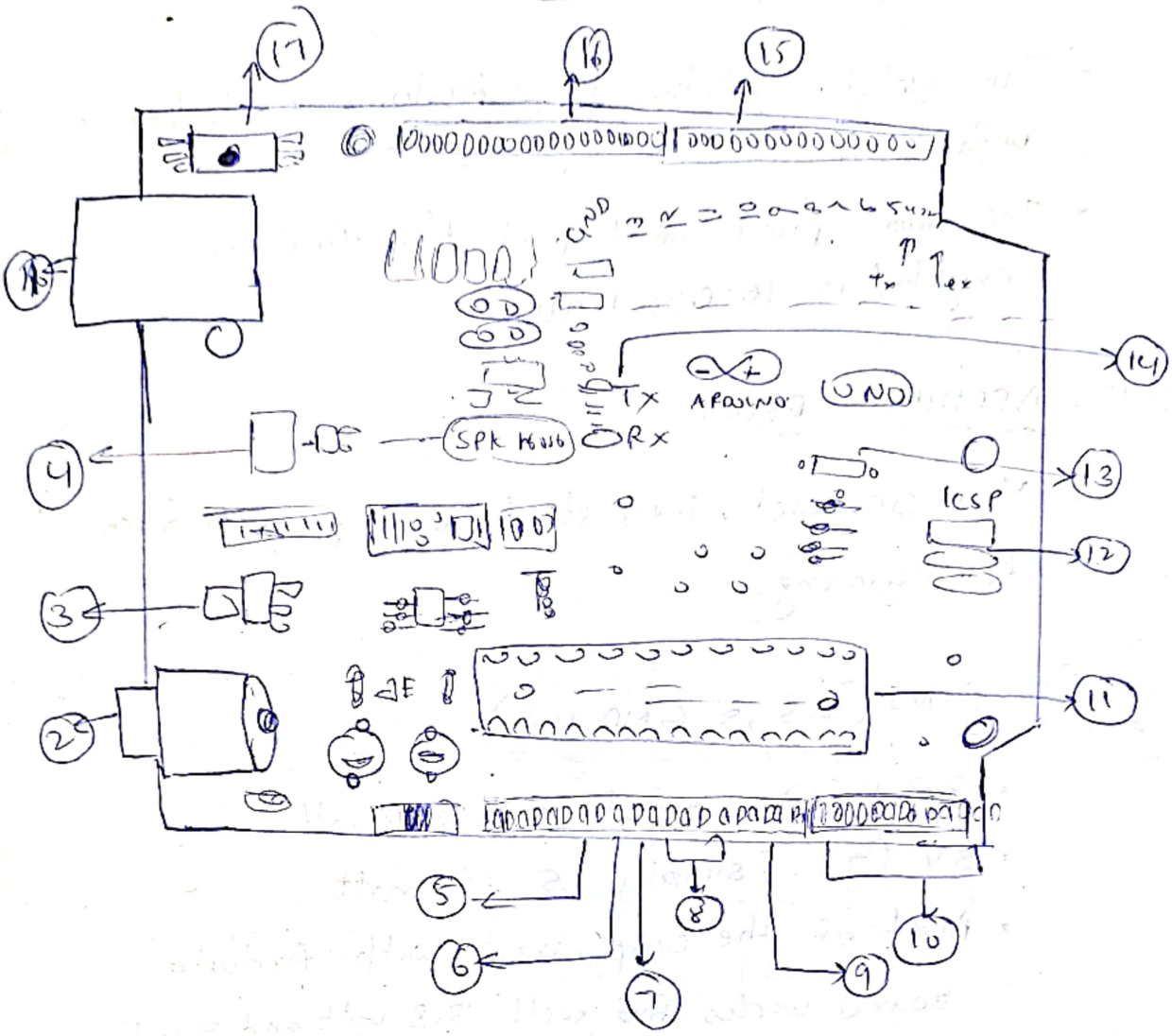
IOT CC

17/12/25



(1) POWER USB

- Arduino board can be powered by using the USB cable from your computer.

(2) POWER (BARREL JACK)

- Arduino boards can be powered directly from the AC mains power supply by connecting it to the Barrel jack

(3) VOLTAGE REGULATOR

- The func. of the voltage regulator is to control the voltage, given to the Arduino board and stabilize.

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(4) CRYSTAL OSCILLATOR

- The crystal oscillator helps Arduino in dealing with issues.
- The num. printed on top of the Arduino crystal is 16.000 H 9H.

(5,17) ARDUINO RESET

- You can reset, i.e, start your program from the beginning.

(6, 7, 8, 9) PINS (3.3, 5, GND, Vin)

- 3.3 V (6) - supply 3.3 O/P volt
- 5V (7) - supply 5 O/P volt
- Most of the comp. used with Arduino board works fine with 3.3 volt and 5 volt
- GND (8)(Ground)
- Vin (9)

(10) ANALOG PINS

- The analog pins UNO board has six analog inputs pins A0 through A5.

(11) MAIN MICROCONTROLLER

- Each Arduino board has its own microcontroller(11) The main IC(Integrated Circuit) on the Arduino is slightly different from board to board.

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(12) ICSP Pin

- Mostly, ICSP is an AVR, a tiny programming header for the arduino consists of MOST, MISO, SCK, VCC & GND.

(13) Power LED indicator

- This LED should light up when you play your Arduino into a power source to indicate that your board is powered up correctly

(14) TX & RX LEDS

- On your board, you will find 2 labels: TX (transmit) and RX (receive)
  First, at the digital pins 0 & 1.
  Second, the TX & RX LED.

(15) Digital I/O:

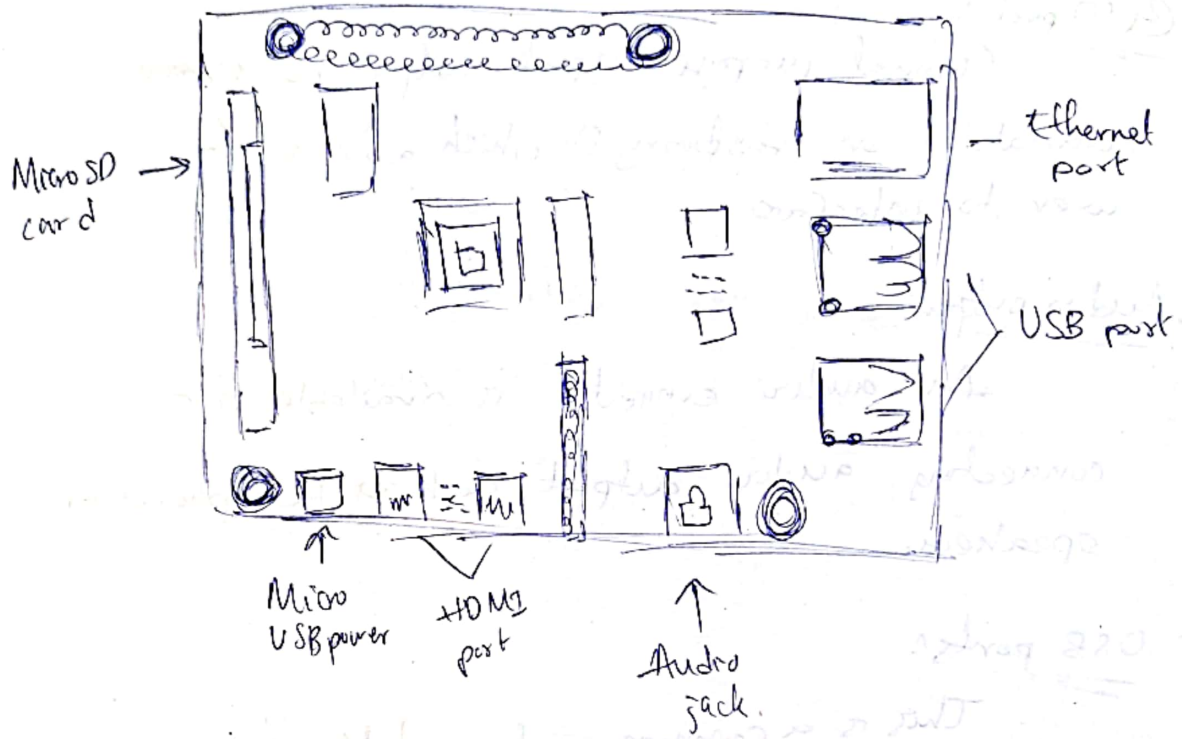
- Arduino UNO board has digital I/O pins which provide PWM output.
- These pins are to read logic values 0 or 1.

(16) AREF:

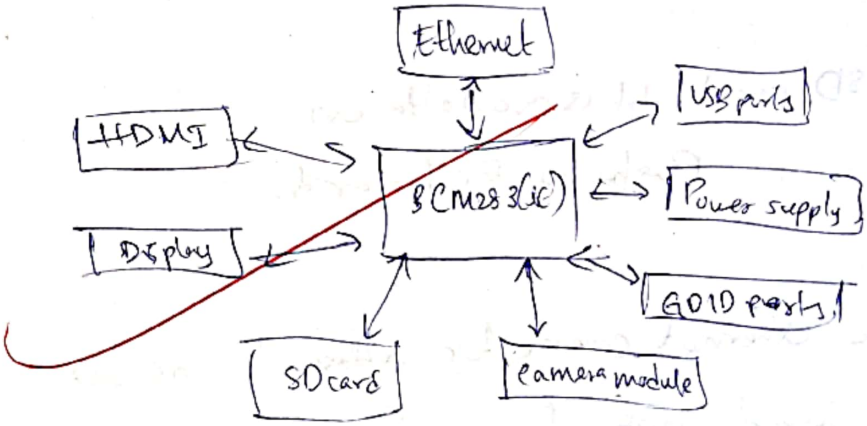
- AREF stands for Analog reference.
- Sometimes, used to set an external reference voltage (b/w 0 & 5).

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Rasberry Pi



Main Blocks of Rasberry Pi :



Majorly Consists :

PICC
ch
Proximity integrated circuit card

Processor :

Rasberry Pi uses Bradcom on chip which is an ARM processor & GPU

HDMI :

High Definition multimedia interface is used for training video or digital audio

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data to a computer or to a digital cable.

GPIO port :
General purpose input output ports are available on raspberry Pi which allows the user to interface.

Audio output:
An audio connector is available for connecting audio output such as headphones & speakers.

USB ports:
This is a common port available for various peripherals such as headphones.

SD card :
The SD card slot is available on Rasberry Pi board.

Ethernet:
The ethernet connector allows to access to the wired network.

Power supply :
A micro USB power connector is available onto which a 5V power supply can be connected.

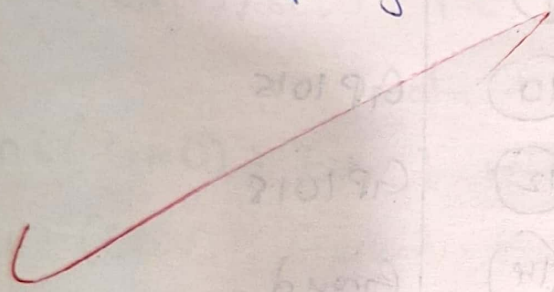
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Camera module :

Camera serial interface (CSI) connects to the Boardcom processor to Pi camera.

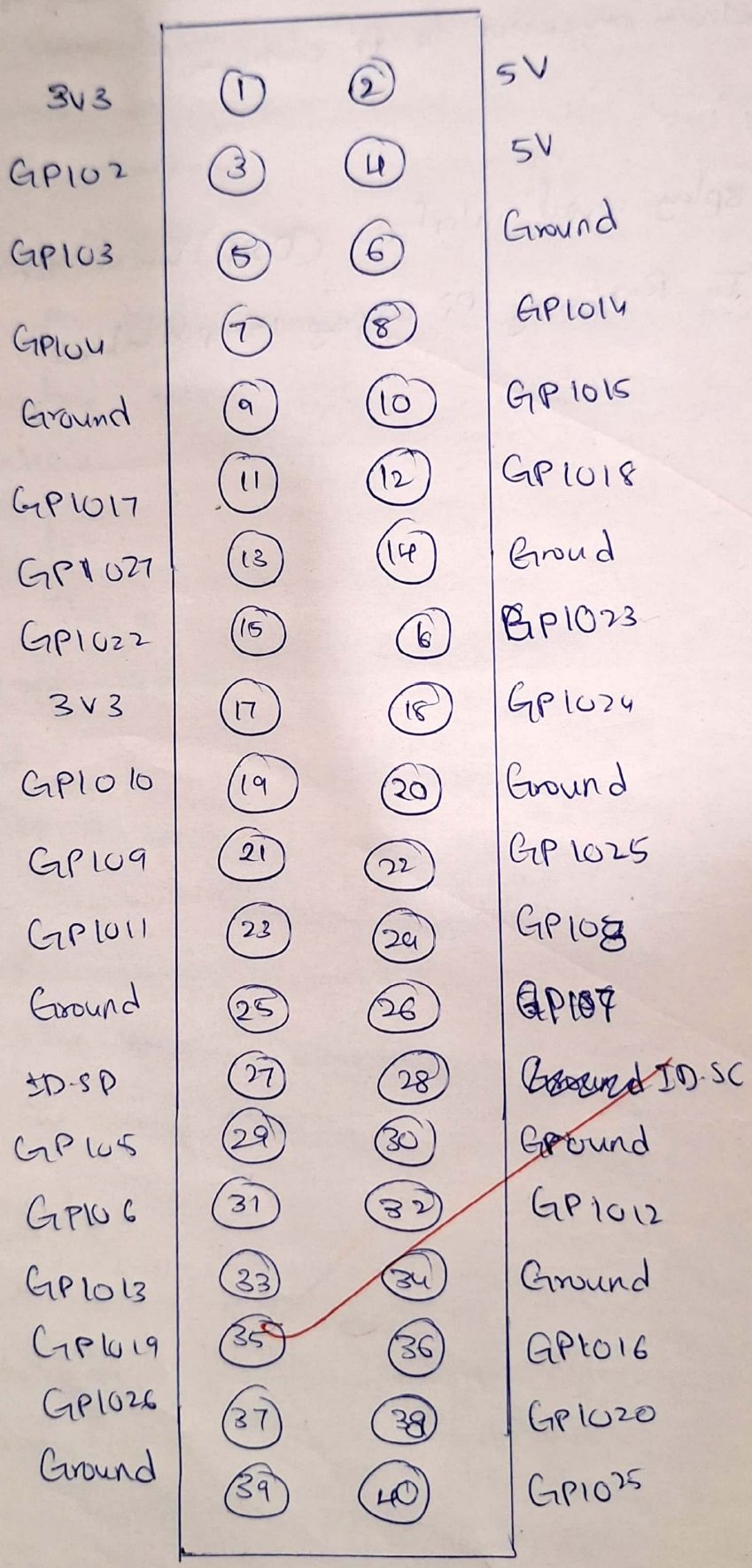
Display :

Display serial interface (DSI) used for connecting LCD, To Raspberry Pi using 15-pin ribbon.



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Rasberry 40 pins & explanation



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19. Pin19 (GPIO25) : General-purpose
19. Pin19 (GPIO20/MOSI) : SPI master out slave in data line.
20. Pin20 (GPIO9/GND) : Ground connection.
21. Pin21 (GPIO9/MISO) : SPI master-in-slave-out details.
22. Pin22 (GPIO25) : General-purpose input/output pin
23. Pin23 (GPIO11/SCLK) : SPI clock signal pin
24. Pin24 (GPIO8/CE0) : SPI chip enable 0 for device selection.
25. Pin25 (GND) : Ground connection.
26. Pin26 (GPIO7/CE1) : SPI chip enable 1 for device selection.
27. Pin27 (GPIO0/ID-SD) : I²C EEP ROM data for ~~HAT~~ identification
28. Pin28 (GPIO1/ID-SC) : I²C EEPROM clock for ~~HAT~~
29. Pin29 (GPIO5) : General purpose.
30. Pin30 (GND) : Ground connection
31. Pin31 (GND) : Ground connection
32. Pin32 (GPIO12/PWM) : PWM-capable output pin
33. Pin33 (GPIO13/PWM) : PWM-capable output pin
34. Pin34 (GND) : Ground connection
35. Pin35 (GPIO19/PWM) : pwm-capable pin often used for audio.
36. Pin36 (GPIO16) : General-purpose input/out pin
37. Pin37 (GPIO26) : General-purpose input/out pin

1. Pin1 (3V3): Supplies 3.3V Power for sensors and low-power modules.
2. Pin2 (5V): Supplies 5V power directly from the Pi Power input.
3. Pin3 (GPIO2/SDA): I^2C data line for communication with peripherals.
4. Pin4 (5V): Additional 5V power output pin.
5. Pin5 (GPIO3/SCL): I^2C Clock line for synchronized data transfer.
6. Pin6 (GND): Ground refers to electrical circuits.
7. Pin7 (GPIO4): General-Purpose digital input or output pin.
8. Pin8 (-GPIO14/TXD): UART transmit pin for serial communication.
9. Pin9 (GND): Ground Connection.
10. Pin10 (GPIO15/RXD): UART (transmit) receive pin for serial communication
11. Pin11 (GPIO17): General purpose input/output pin
12. Pin12 (GPIO18/PWM): PWM-capable pin for motors or LED dimming.
13. Pin13 (GPIO27): General purpose input/output pin
14. Pin14 (GND): Ground connection.
15. Pin15 (GPIO22): General purpose input/output pin
16. Pin16 (GPIO23): General purpose input/output pin
17. Pin17 (3V3): Additional 3.3V Power output pin
18. Pin18 (GPIO24): General purpose input/output pin

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38. Pin38 (GPIO20): General purpose

39. Pin39 (GND): Ground connection.

40. Pin40 (GPIO27): General purpose input output pin

